Ram Moment Coordinates and the Spatial Conversion of Disk Transfer Mass

A source-specific dependency audit of manuscript M460

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Abstract

The pinned manuscript claims critical honeycomb bridge mass comparable to $R^{-1/4}$, fixed-corridor and unrestricted first moments $R^{13/12+o(1)}$, and the resulting annealed mean-length exponent 4/3. We audit the complete local source from the finite exploration through the rapidity orbit, disk interpolation, reflected-current estimate, determinant normalization, polygon localization, and recoverable two-arm sewing. The complex moment matrix is retained as $G+\mathrm{i}J$, with its physical metric and evaluation covector transported together; the generic Cauchy inverse is not continued as a matrix at coincident rapidities. We prove exact moment-coordinate, effective-Gram, alternating spectral, and finite-residual certificates and a conditional sewing theorem that retains the actual arm normalization, halves the displayed arm-loss exponent, and isolates the shared-offset summability threshold. These are local verification and conversion gains, not an independent verification of the manuscript's asymptotic theorems, a fixed-length displacement theorem, or a historical novelty claim. Contribution assessment: C3/5, exact effective-Gram and alternating physical-metric certificates and a sharper conditional sewing account are realized while the source contour and spatial inputs remain independently unverified.

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1 Pinned problem and scope

1.1 Identity of the source

This report concerns catalogue ID 460, family 237, at the supplied commit

`adc7f1241b42e322a6451854ab7e4b4c146bf78a`.

The title is *Disk transfer representations and confined bridge mass*, dated September 26, 2026. The local source prefix, relative to `C:/Users/mural/OneDrive/Documents/OpenAI-722-math/`, is

`_sources/openai-math/preprints/Disk-transfer-representations-and-confined-bridge-mass-September-26-2026/`.

Subsequent source locators are relative to this prefix. The catalogue row agrees with the supplied pin. The local source directory is not a Git checkout, so no claim of a successful local `git rev-parse` is made; the input ledger records actual file hashes instead. Applicability L5 measures direct relevance, not correctness or realized contribution.

The complete files `build/sections/introduction.tex`, `boundary.tex`, `analytic.tex`, `geometry.tex`, and `visits.tex` in that section directory were read. The last is included inside `geometry.tex`, not directly by the main file. The main file, references, README, and four figure sources were also read. There is no separate mathematical appendix in this source tree; the long stability and normalization arguments inside `analytic.tex` were included in the read scope. Reading is not an independent proof check of every identity.

We distinguish four statuses throughout: *source claim* means a statement supplied and argued by the manuscript; *local theorem* means proved here; *conditional consequence* means proved here from explicit source inputs; *unreplayed dependency* means read but not independently established in this report. None of the latter is silently promoted by exact coordinate calculations.

1.2 Lattice, domains, weights, and endpoint conventions

Set

$$e_2 = (0,-1), \quad e_0 = (\sqrt{3}/2, 1/2), \quad e_1 = (-\sqrt{3}/2, 1/2), \quad U = e_0 - e_1, \quad V = e_0 - e_2.$$

The two vertex classes are $jU+kV$ and $jU+kV+e_0$ for $j,k \in \mathbb{Z}$. Edges have unit length. Admissible walls are $e_i \cdot y = t$ with $t \in \frac{1}{2} + \frac{3}{2}\mathbb{Z}$; they cut edges at midpoints without hitting vertices. A port is

such a midpoint. A path visits distinct lattice vertices strictly inside its domain and includes its two terminal half-edges. With ℓ the vertex count,

$$x = (2+\sqrt{2})^{-1/2} = (2\cos(\pi/8))^{-1}, \qquad w(\gamma) = x^{\ell(\gamma)}.$$

Terminal half-edges carry no additional weight. This convention differs by a factor of x from edge-count weights on a nonempty vertex-to-vertex path. We never interchange these normalizations inside sewing formulas.

For a strip between parallel admissible walls at distance $R \in \frac{3}{2}\mathbb{Z}_{>0}$, fix one port a on the lower wall and sum the terminal port over the upper wall. A bridge stays strictly inside the strip apart from the ports. For a unit tangent τ, put $\xi(y) = \tau \cdot (y-a)$ and define

$$B(R) = \sum_{\gamma} w(\gamma), \qquad M(R) = \sum_{\gamma} w(\gamma)\ell(\gamma).$$

The subscript C restricts both sums to $\sup_{y\in\gamma} |\xi(y)| \le CR$, including the ports. All finite lengths are summed. The probability is w/B or w/B_C, not a uniform fixed-length probability and not a two-fixed-endpoint law.

The source's main claim has the quantifier order

$$\exists C_0 < \infty \quad \forall \text{ fixed } C \ge C_0 : \quad B_C(R) \asymp B(R) \asymp R^{-1/4}, \quad M_C(R), M(R) = R^{13/12+o(1)}. \tag{S1}$$

The constants in $\asymp$ may depend on fixed C but not R. The corridor constant is independent of exponent slack. $f(R) = R^{a+o(1)}$ means $\log f(R)/\log R \to a$ along admissible scales; it does not assert a limiting amplitude.

For half-plane arches, one port is fixed and the other is $m > 0$ port spacings to the right; the corresponding positive mass is $h(m)$. With both directions summed let μ be the unnormalized measure. The further source claims are

$$h(m) \asymp m^{-5/4}, \quad h(m+1) \le h(m), \quad \mu(H > R) \asymp \mu(\text{diam} > R) \asymp R^{-1/4}. \tag{S2}$$

Conditioning on $R \le \text{diam} \le CR$, or on $H \ge R$ and diam $\le CR$, for sufficiently large fixed C, is claimed to give mean length $R^{4/3+o(1)}$. For the admissible triangle

$$D_R = \{y : e_i \cdot y < R \ (i = 0,1,2)\}, \quad I_R = \{v : e_i \cdot v < (1-\varepsilon)R \ (i = 0,1,2)\},$$

where $R \in \frac{1}{2} + \frac{3}{2}\mathbb{Z}$ and small fixed $\varepsilon > 0$, ordered chord endpoints are both summed and the law is normalized on chords visiting I_R. The source claims uniformly for $v \in I_R$ a visit probability $R^{-2/3+o(1)}$ and mean length $R^{4/3+o(1)}$.

2 Full-source dependency audit

The following audit follows the actual logical order. It is a dependency account, not a replacement proof of the manuscript. Source labels are literal labels in the files, enabling targeted review without relying on the screening pack or a routing map.

2.1 Exploration, transfer vacuum, and boundary inverse

B1: positive finite exploration. In `build/sections/boundary.tex`, `c2:eq-1`–`c2:eq-2`, the spin is $s = 3/8$, $\lambda = \pi/8$, and $c = 2\cos(3\pi/8)$. The two fresh-vertex weights $xe^{\pm is\pi/3}$ sum to one. Stopping before a repeated vertex avoids charging that vertex twice. The two exterior loop traversals

add turns $\pm 4\pi/3$ and cancel. For a convex admissible domain the remaining exit turn satisfies $|W_{ab}| \le \pi$, whence

$$\sum_b e^{\mathrm{i}sW_{ab}} h_D(a,b) = 1, \qquad c \sum_{m>0} h(m) \le 1.$$

Finiteness here is available before any exponent calculation and supplies the componentwise transfer bounds. Positivity is used on real homogeneous weights; it cannot be assumed for the deformed rapidities.

B2: a physical analytic vacuum. The local $\check{R}(u)$ has the five nonvacant connectivity coefficients A, B, C, D, E divided by T_u. The source enumerates the noncrossing three-box connectivities to prove the braid relation `c2:eq-3`, with closed loops assigned zero. Its homogeneous row has block form

$$\mathcal{T}(0) = \begin{pmatrix} 1 & 0 \\ g & K \end{pmatrix}.$$

Finite-stack masses $\sum_{j<h} K^j g$ are bounded by products of finite half-plane arch masses. Accessibility of every nonempty state then implies entrywise convergence of $\sum_{j\ge 0} K^j$, invertibility of $I - K$, and uniqueness of the empty-normalized fixed vector $(1, (I-K)^{-1}g)$. This is the source's justification that the vector counts actual arcs, rather than merely solving exchange equations. The annular state retains an arc's lift interval of length less than the period. Forgetting those lift data occurs only later for disk states.

B3: orbit algebra. For fixed $N \ge 3$, distinct nearby rapidities define $x_i = e^{2\mathrm{i}\eta_i}$, $q = e^{\mathrm{i}\pi/4}$, and $\mathcal{C}_{ij} = (x_i + q x_j)^{-1}$. These x_i are not the critical vertex weight x. The endpoint relations `c2:eq-4` lead to Hecke operators g_l, a weighted rotation Ω, and commuting transports Y_i on the $N!$-point permutation orbit. The source proves a one-coordinate reduction `c2:eq-7`; a positive limiting moment pencil supplies simple positive limiting eigenvalues and cyclicity. The sign-change argument forces each limiting eigenpolynomial to have degree $N-1$ and excludes vanishing evaluation at i. Nonzero adjacent eigenspace connections then give the full orbit and a scalar commutant. These are nontrivial source dependencies, not consequences of naming the coordinates Ram moments.

B4: only the scalar continues. The telescoping endpoint sum and two polynomial resolvents give

$$\sum_i x_i - c \sum_{i \ne j} e^{\mathrm{i}(\eta_i + \eta_j)} H_{ij}(\boldsymbol{\eta}) = \mathbf{1}^t (\mathcal{C}^t + q\mathcal{C})^{-1} \mathbf{1}. \tag{1}$$

This is `prop:boundary-inverse`, `c2:eq-9`. The identity is first obtained on a nearby open set with simple spectrum, not by an identity theorem from one isolated progression alone. Meromorphic continuation and the analytic left side provide a removable *scalar* singularity. At coincident rapidities the original Cauchy matrix has repeated rows. Substituting equal rapidities into that matrix and then inverting is invalid.

B5: confluence and the evaluation norm. Divided-difference row and column functions converge, for fixed N, to monomials. The evaluation vectors transform at the same time. The result

is the source equation `c2:eq-10`:

$$N\left(1-c\sum_{m=1}^{N-1}h_N(m)\right)=2w^*\mathfrak{M}_N^{-1}w, \qquad w_j=\mathrm{i}^j,\quad 0\le j<N, \tag{2}$$

$$(\mathfrak{M}_N)_{jk}=\int_{\mathbb{R}}t^{j+k}\rho(t)\,dt, \qquad \rho(t)=\frac{e^{\lambda t-\mathrm{i}\lambda}+e^{-\lambda t+\mathrm{i}\lambda}}{2\cosh(\pi t/2)}. \tag{3}$$

Writing $\mathfrak{M}_N=G_N+\mathrm{i}J_N$ gives

$$g(t)=\Re\rho(t)=\frac{\cos\lambda\cosh(\lambda t)}{\cosh(4\lambda t)}, \qquad \frac{\Im\rho(t)}{g(t)}=-\tan\lambda\tanh(\lambda t).$$

Thus $G_N>0$ and $-\tan\lambda G_N\preceq J_N\preceq\tan\lambda G_N$. The source compares the real inverse pairing with $w^*G_N^{-1}w$, then compares $g(t)$ with $\operatorname{sech}(3\pi t/8)$. The rescaling $v=3t/8$ changes evaluation to $3\mathrm{i}/8$. For the mass-one measure $dv/\cosh(\pi v)$ the generating function

$$(1-\mathrm{i}z)^{-1/2+\mathrm{i}v}(1+\mathrm{i}z)^{-1/2-\mathrm{i}v}=\sum_{m\ge0}p_m(v)z^m$$

has bilinear generating-function pairing $(1-zz')^{-1}$. Consequently the squared evaluation norm is $\sum_{m<N}|p_m(3\mathrm{i}/8)|^2\asymp N^{3/4}$, using the positive coefficient convolution of $(1-y)^{-3/4}(1-y^2)^{-1/8}$. The factor N^{-1} in (2) is essential for the deficit $N^{-1/4}$ in `c2:eq-11`.

B6: a separate geometric boundary conversion. Lost projected arches imply an upper endpoint-displacement tail and a lower horizontal-width tail, not immediately a height law. The two wedge identities `c2:eq-13` compare tails of $L_\alpha=\max(y\cot\alpha-z)$ for $\alpha=\pi/3,2\pi/3$. Contraction gives upper tails; averaged differences and absorption give the lower height and endpoint tails `c2:eq-14`. Exhaustion of the strip then gives $B(T)=c\mu_+(H>T)$. A distinct two-exploration switch proves monotonicity of $h(m)$; the tail alone does not prove the pointwise $m^{-5/4}$ estimate. The quantitative unfolding subsection records $c_nx^n\le\exp(C\sqrt{n}\log(n+1))$ with explicit endpoint conversion. It is not used here as a proof of a fixed-length 3/4 displacement law.

2.2 Disk interpolation and the analytic cylinder estimate

All locators in this subsection belong to `build/sections/analytic.tex`.

A1: faithful disk realization. Partial noncrossing pairings of N ports span $\mathcal{D}_N$. In the three-state spin realization $0,+1,-1$, the cup is $\mathsf{c}_{st}=q^s\delta_{s,-t}$. Caps are transposes, not Hermitian adjoints: $\mathsf{c}^t\mathsf{c}=q^2+q^{-2}=0$ removes loops and the snake identity straightens strands. A Dyck-word triangularity argument proves faithfulness. The source checks unitarity and both fusion maps J,K at special rapidities in `c2:eq-20`–`c2:eq-21`. This bilinear diagram contraction is not a positive physical Gram metric.

A2: scalar normalization is fixed, not adjustable. The Pfaffian polynomial p_N has individual degree at most $N-1$; its fusion/deletion formulas are `c2:eq-22`. Proposition `c2:prop-disk-vector` proves

$$\Phi_{N,\alpha}=\left(\prod_i x_i^{-a_i/2}\right)p_N\psi_{N,\alpha}, \qquad \deg_{x_i}\Phi_{N,\alpha}\le N-1-a_i.$$

Its proof includes shared-endpoint and chain compatibility, the reverse chain checks at sizes three and four, independence of the chosen occupied interpolation label, and transfer invariance. The empty component one fixes normalization. Ignoring one of these compatibility checks would leave an existence gap in the interpolated vector. The two-site cluster limit is needed separately in scalar uniqueness at balanced sizes.

A3: physical gluing and algebraic gas are distinct. At balanced homogeneity $N = 2n$, the twisted bilinear pairing $V_{n,n}$ counts unrooted, unoriented, vertex-disjoint polygons separating two antipodal seam face centers on the cylinder of period $2nU$. Each loop has weight $2\cos\theta\, x^{|\gamma|}$. Nonseparators cancel by the spin factors, whereas separators sum two orientations. There are at most n separators, so this is a polynomial in $2\cos\theta$ with nonnegative coefficients, and $V_{n,n}(\pi/2) = 1$. The interpretation is not claimed for arbitrary rapidities. Scalar uniqueness `c2:lem-scalar-uniqueness` and the explicit gas identity `c2:eq-24` are proved using degree bounds, pole cancellation, fusion and cluster limits. Gas labels are not the disk spin labels. Residues repeat no site; the original homogeneous gas has at most $2n$ letters.

A4: current reflection is an infinite analytic dependency. The auxiliary period T is not the physical circumference. Oscillator traces require energy damping, finite-shell zero-mode calculations, momentum-lattice parity, and the Weyl vacuum sign. The reflection changes the interaction from M to M^{-1}, as well as replacing the source by

$$G(v) = \frac{\cos(\nu v) - \cos(\pi/6)}{\cos(\nu v) + \cos(\pi/6)}, \qquad \nu = \frac{\pi}{3\lambda}.$$

The trace initially exists at nearby contours; remote-contour identities are obtained by continuation of scalar expressions, not by asserting a trace-class operator identity everywhere. These distinctions in `c2:sec-currents` matter to any independent verification.

A5: residue stability and uniform summation. Section `c2:sec-stability` lists center compounds $\alpha_0 - \sum_{t\in J} \alpha_t$, outer negative letters, and the across pair $-\alpha_{-3/2} - \alpha_{3/2}$. At least $N/6$ elementary-letter units are represented by good, source-suppressed particles. The cross-label term is positive only *after integration* against the signed row measures, not frequency by frequency. The complementary form is coercive after fixing a small contour separation; its regular zero mode is controlled on $Q_s = 0$. Smearing is applied to whole compounds so that total coincident-log coefficients are nonnegative. The loss is $O(N\log(B + N + 2))$, $B = \Re T$. The local bin and source costs yield summable suppression $-cN^2/\log(N + 2)$ and a full particle cost $e^{o(B)}$ when $B = (2/\nu)\log n$. Bent-contour comparison consumes spare real coercivity; it does not make the complex energy positive. Holomorphic dependence then permits a maximum principle at nome zero. The unreplayed result is

$$\frac{\log|\mathcal{Z}_n(\theta)e^{-A_\infty n^2}|}{\log n} \le \frac{3}{8} - \frac{2}{3\pi^2}\Re\theta^2 + o(1), \quad A_\infty = \log\frac{4(2+\sqrt{2})^2}{81}.$$

Finite matrix tests cannot certify this uniform complex-neighborhood estimate or its infinite-order convergence.

A6: normalization, compression, and saturation. The independent Pfaffian normalization `c2:eq-relative-moment` uses two confluent moment determinants. Translating the numerator weight by i/3 crosses no pole. The resulting multiplication ratio $S(t)$ satisfies $\Re S \ge \delta > 0$, $S(t) \to 1$

exponentially. For the polynomial projection P, source equation `c2:eq-V9` gives local trace density $3\lambda \log n/\pi^2$ and $\|(1-P)bP\|_{\mathrm{HS}} = o(\sqrt{\log n})$. The latter is an omitted-component estimate; exact finite compression alone would not imply it. Accretive logarithmic differentiation then gives the power $5/24$ in P_{2n}^2. The exponential factors cancel and $3/8 - 5/24 = 1/6$. The harmonic upper function

$$h(\theta) = \frac{1}{6} - \frac{2\Re\theta^2}{3\pi^2}$$

is attained at $\theta = \pi/2$ because the fugacity-zero partition equals one. The source's subharmonic barrier lemma propagates saturation to zero using positivity and monotonicity on the real interval. The final source conclusion is `c2:prop-cylinder-exponent`, $V_{n,n}(0) = n^{1/6+o(1)}$, not a result independently proved here.

2.3 Polygon localization, visits, and exterior arms

The following locators are in `build/sections/geometry.tex`, except the visit recursion in `build/sections/visits.tex`.

G1: two adjacencies and translation-class mass. With $p = 1/4$, the boundary input gives $Q(n) = \sum_{d\ge 1} \min(d,n)h(d) \asymp n^{1-p}$. The vacancy identity `c2:eq-B1` has signed individual losses. Only their interval sums have positive comparison `c2:eq-B2`. One-wall adjacency gives $\sum_k f_{1,k} \lesssim B^2/Q$; the four-vacancy telescoping and absorption give $f_{1,1} \lesssim R^{-3}$. The offset between top and bottom lists is summed throughout. Cutting a polygon at leftmost extreme vertices is an injective translation-class encoding with weight change x^2 and $O(L)$ possible heights. Therefore its dyadic shape mass is $O(L^{-2})$. Neither the individual signed loss nor a rooted polygon count can be substituted for these objects.

G2: winding loops and planar nesting. For a finite strip on the cylinder the source constructs a nonnegative matrix $A_{\gamma\eta} = \sqrt{2} w_\eta \mathbf{1}_{\eta \not\preccurlyeq \gamma}$. It proves $(I-A)S > 0$ without asserting $S > 0$, excludes spectral radius one by a nonnegative left Perron vector, and uses a top-row deformation and continuity to infer $\rho(A) < 1$. This controls winding separator mass by $2/\sqrt{2}$ after exhaustion. Large contractible loops cost $O(n/L)$ per dyad by shape mass and $O(nL)$ placements. Deleting their avoidance constraints costs a bounded exponential factor. Thus `eq:disk-localization` reads

$$Z_{cn}(f)^2 \le V_{n,n}(0) \le e^C Z_{cn}(f)^2,$$

where Z_R counts disjoint nested polygons in a radius-R ball, with factor two per loop. This is the source of $z = 1/12$, not an algebraic square-root operation without spatial comparison.

G3: finite visit recursion. For a finite filled vertex set D, the spin-weighted ordered chord mass $T(v,D)$ is real but its individual terms need not be positive. Proposition `prop:visits` proves the terminating recursion

$$\begin{aligned} T(v,D) = 3 &+ \sum_{\Gamma \ni v} 2x^{|\Gamma|-1}[-\cos(s(2\pi - \tau_v(\Gamma)))] \\ &+ \sum_{v \in \Gamma^\circ} 2x^{|\Gamma|} T(v, \Gamma^\circ). \end{aligned} \tag{4}$$

The omitted starting vertex explains $x^{|\Gamma|-1}$. The two coefficients are positive. Unrolling strict interiors gives monotonicity on filled sets and comparison with the sum of the three incident-face

nesting partitions. For convex admissible domains the positive ordered visit mass $A_D(v)$ satisfies

$$\cos(3\pi/8)A_D(v) \le T(v,D) \le A_D(v).$$

This step does not select the terminal wall. Treating T as already bridge-only would bypass the central difficulty of the manuscript.

G4: triangle and arch means. For interior v, $A_D(v) = R^{z+o(1)}$. The macroscopic triangle chord normalizer is R^{1-p}, so normalized visits have exponent $z + p - 1 = -2/3$; summing R^2 vertices gives $1 + p + z = 4/3$. The arch lower bound adds one exterior arm and varies three walls. For weighted representation multiplicity $m(\xi)$, the source obtains first moment $R^{3+z-p-o(1)}$ and second moment $R^{5+z-2p+o(1)}$. Cauchy–Schwarz gives distinct mass $R^{1+z-o(1)}$ through a bulk vertex over wall locations. Summing vertices and dividing by the number of source-wall pairs gives the same per-source first-moment exponent. The upper bound uses translation averaging into one triangle, not multiplicity assumptions.

G5: separated endpoint supply. The marked three-leg identity `c2:eq-K1` uses the derivative in a vertex fugacity. The intermediate quantities P and $\widetilde{P}$ are not asserted equal individually; their specific three-term combinations agree after marked tripod cancellation. For $0 < \tau < 1 - p$ and $G_{ab} = (1 + |b - a|)^{-\tau}$, the source proves

$$\left\langle \sum_{a,b} G_{ab} K_{ab} \right\rangle \le H^{z-\tau+o(1)}.$$

Here brackets average three independent wall-offset windows of order H. The deletion loss uses filled-domain monotonicity and an inward wall telescoping gain $O(L/H)$. The same-side diameter estimate sews an unrestricted completing arch; its polygon may be much larger than L, so all diameter dyads above L must be summed. Consequently, for every fixed $\epsilon > 0$, endpoints separated by at least $H^{1-\epsilon}$ retain average mass $H^{z+o(1)}$. There is no pointwise-in-triangle or fixed-endpoint conclusion here.

G6: tubes, bends, and recoverable multiplicity. The source first proves fixed-end bridge supremum and total variation $O(L^{-1-p})$, plus lateral tightness $O((KL)^{-p})$. A plateau and varying subslab cuts supply narrow straight tubes with fixed-end mass $\gtrsim L^{-1-p}$. Widening from $g = R^{1-\epsilon}$ uses $O(1 + \epsilon \log R)$ geometric slabs, giving $R^{-1-p-O(\epsilon)}$. At most two bends are made with small triangles; two cut vectors recover common bend chords, gap bridges, and fixed-end straight pieces. Different-side and same-side central chord layouts give disjoint exterior arms in a fixed corridor. Each arm family and its thinning factor depend only on its own initial port, triangle, and role, not the other central endpoint. That last condition is essential when two representations select their exterior entry and exterior exit from different triangles.

G7: counting actual bridges. For fixed ϵ, the source retains chord mass $R^{3+z-o(1)}$ over three offset windows. Its displayed two-arm moments are at least $R^{3+z-2p-2K_{\mathrm{loss}}\epsilon-o(1)}$ and at most $R^{6+z-4p+o(1)}$, respectively. Same-side offsets require summing $(1 + |d - d'|)^{-2p}$, hence $2p < 1$ at this step. The resulting distinct mass through v is $R^{z-4K_{\mathrm{loss}}\epsilon-o(1)}$. Summing R^2 vertices and dividing by $O(R)$ source ports yields $M_C(R) \ge R^{1+z-o(1)}$, with the corridor fixed before $\epsilon \downarrow 0$. Section 6 strengthens this local bookkeeping conditionally without replacing the geometric recovery proof.

For the unrestricted upper bound, polygon shapes and strip-restricted placements bound $A_R(v) \le R^{z+o(1)}$ with all starting ports summed. The source uses the exact translation identity

$$M(R) = \sum_{v \in \mathcal{F}_R} A_R(v), \qquad |\mathcal{F}_R| = O(R),$$

where $\mathcal{F}_R$ is a vertex fundamental set modulo U. All sums are nonnegative, so exhaustion and reindexing use monotone convergence. Finally the separate corridor mass tightness gives $B_C \asymp B$. Division by the correct mass yields $1 + z + p = 13/12 + 1/4 = 4/3$.

3 Actual private Ram correspondences

3.1 Original passages read locally

The following originals, not only catalogue routing maps, were read. The paths are confidential local provenance, not public literature queries.

1. `C:/Users/mural/OneDrive/Documents/RamFoundationTRs/TR-rev-1-old-tr250/TR-rev-1-old-tr250.tex`: "Physical spaces precede matrices," "What a Ram representation means in this report" (`eq:moments`), and "A concrete polynomial Ram construction." The first two occur at lines 121–181 and the polynomial construction at 455–477 in the read version.

2. `C:/Users/mural/OneDrive/Documents/RamCalBooks/volumes/volume21/chapters/ch01d-physical-metrics.tex`: the entire chapter file, including the following literal labels:

   ```
   v21c01:physical-gram-definition
   v21c01:metric-symmetry
   v21c01:residual-decomposition
   v21c01:gram-equivalence.
   ```

3. `C:/Users/mural/OneDrive/Documents/RamFoundationTRs/TR-rev-1-old-tr117/TR-rev-1-old-tr117.tex`: "Basic Ram Transform and Ram Localization," including `eq:RT`, `eq:RLT`, `eq:IRLT`, and "Equivalence of global and Ram forms," lines 238–315 in the read version.

TR250 explicitly defines moments $m = Ec$ with $E_{ij} = \ell_i(\phi_j)$ on a selected finite physical space. An invertible E changes coordinates, not that space, and does not create a new PDE theorem. Its positive polynomial moment construction is directly relevant to the polynomial space behind (2); its diffusion boundary conditions are not transferred to honeycomb walks.

Volume 21 gives $G = J^*J$ for injective synthesis, the metric symmetry $GB = B^*G$, and the omitted residual $\|(I - P)AJc\|$. Here the synthesis is polynomial reconstruction in $L^2(g(t)dt)$. The compressed multiplier is the bounded real function $a(t) = -\tan\lambda \tanh(\lambda t)$. Its compression is self-adjoint in the physical metric; $I + \mathrm{i}a$ is accretive, not self-adjoint. For the separate determinant normalization, the source's $(1 - P)bP$ estimate is precisely the kind of missing component the private chapter says must not be erased. A finite zero residual does not prove that Hilbert–Schmidt estimate.

TR117's RLT/IRLT changes a kernel's source/displacement coordinates. It does not invert the integral operator. For an integrable kernel $k'(u, t)$ on $[r, s]$, setting $k(t, \alpha) = k'(t + \alpha, t)$ and $\beta = u - t$ gives

$$\int_r^s k'(u, t) f(t)\, dt = \int_{u-s}^{u-r} k(u - \beta, \beta) f(u - \beta)\, d\beta.$$

This is an exact representation identity when the integrals exist. There is no valid inference from it to a cylinder inverse, a bridge exponent, or an independent probability law. We use the finite moment interpretation rather than an unjustified continuum inverse.

3.2 Exact moments and coordinate transport

Put $H = L^2(\mathbb{R}, g(t)dt)$, with inner product conjugate-linear in the first slot. Let $X_N = \mathcal{P}_{N-1}$, polynomials of degree less than N, and $J_N c = \sum_{j=0}^{N-1} c_j t^j$. All polynomial moments are finite because $g(t)$ decays like $e^{-3\lambda|t|}$. Evaluation at a nonreal point is a bounded functional on this finite space; it is not a bounded point-evaluation functional on all of H.

Proposition 3.1 (Exact finite Ram moment bank)**.** *Let $\ell_j(f) = \int_{\mathbb{R}} t^j f(t)g(t)dt$ for $0 \le j < N$. Then $m = G_N c$ is an invertible moment bank on X_N. In moment coordinates its physical metric, complex form matrix, and right covector for (2) are respectively*

$$G_N^{-1}, \qquad G_N^{-1}\mathfrak{M}_N G_N^{-1}, \qquad G_N^{-1} w.$$

For every invertible complex S under $c = S\widetilde{c}$, the same three objects transform by congruence and dual transport:

$$\widetilde{G} = S^* G_N S, \quad \widetilde{\mathfrak{M}} = S^* \mathfrak{M}_N S, \quad \widetilde{w} = S^* w, \qquad \widetilde{w}^* \widetilde{\mathfrak{M}}^{-1} \widetilde{w} = w^* \mathfrak{M}_N^{-1} w.$$

Proof. For $c \neq 0$, $J_N c$ is a nonzero polynomial, so $c^* G_N c = \int |J_N c|^2 g > 0$. Thus G_N is invertible. Substitute $c = G_N^{-1} m$ in the norm and sesquilinear form. For the final statement substitute S into both test and trial coordinates and use $(S^*\mathfrak{M}_N S)^{-1} = S^{-1}\mathfrak{M}_N^{-1}(S^*)^{-1}$. All covectors must be moved. □

The scalar $w^* c$ is evaluation of $J_N c$ at $-\mathrm{i}$; its conjugate evaluation vector at i is w. Because G_N has real entries, the two positive evaluation norms agree. This convention prevents a transpose from being silently replaced by an adjoint in the source's original bilinear confluence. Equation (3.1) applies after the complex moment form has been identified, not to an arbitrary independent change of the generic Cauchy rows and columns.

Moments can be generated exactly without truncating the real line. From the secant transform, for $|\Re z| < 3\lambda$,

$$F(z) := \int_{\mathbb{R}} e^{zt}\rho(t)dt = e^{-\mathrm{i}\lambda}\sec(z+\lambda) + e^{\mathrm{i}\lambda}\sec(z-\lambda), \qquad (\mathfrak{M}_N)_{jk} = F^{(j+k)}(0). \tag{5}$$

Differentiation under the integral is justified on smaller strips by exponential domination. If $Q_0(u) = 1$ and $Q_{n+1}(u) = uQ_n(u) + (1+u^2)Q_n'(u)$, then $\partial_z^n \sec z = \sec z\, Q_n(\tan z)$ by induction. This gives a finite symbolic recurrence at $\lambda = \pi/8$. The real weight is even and the imaginary weight odd; hence $G_{jk} = 0$ when $j+k$ is odd and $J_{jk} = 0$ when $j+k$ is even. These parity zeros are exact diagnostics, not fitted numerical patterns. More explicitly, with $\tau = \tan\lambda = \sqrt{2} - 1$ the moments are

$$F^{(k)}(0) = \begin{cases} 2Q_k(\tau), & k \text{ even}, \\ -2\mathrm{i}\tau Q_k(\tau), & k \text{ odd}. \end{cases} \tag{6}$$

Indeed Q_k has parity k, $\sec(-\lambda) = \sec\lambda$, and $\sec\lambda\cos\lambda = 1$. Thus the physical moment entries lie in the exact field $\mathbb{Q}(\sqrt{2}, \mathrm{i})$, with no transcendental rounding needed for fixed N.

4 Effective Gram and alternating scalar certificates

This section records the collective audit's independently checked finite-matrix refinements with all-N proofs. Their algebraic validity is independent of the source's walk interpretation. Interpreting $2Q/N$ as a physical boundary deficit remains conditional on (2), established by the source's earlier transfer/orbit arguments for $N \geq 3$.

Theorem 4.1 (A positive effective Gram matrix)**.** *Let $G > 0$ and $J = J^*$ be finite Hermitian matrices, let $A = G^{-1/2} J G^{-1/2}$, and suppose $\|A\| \leq \tau$. Put $M = G + \mathrm{i}J$ and*

$$H = G + JG^{-1}J, \qquad Q = \Re(w^* M^{-1} w), \qquad Q_0 = w^* G^{-1} w.$$

The Hermitian part of M^{-1} is exactly H^{-1}, so

$$Q = w^* H^{-1} w, \qquad G \preceq H \preceq (1+\tau^2) G. \tag{7}$$

For any trial vector v, set $r_H = w - Hv$ and $L(v) = 2\Re(w^* v) - v^* H v$. Then

$$Q - L(v) = r_H^* H^{-1} r_H, \quad \frac{\|r_H\|_{G^{-1}}^2}{1+\tau^2} \leq Q - L(v) \leq \|r_H\|_{G^{-1}}^2. \tag{8}$$

In particular $Q = \sup_v L(v)$ with unique maximizer $H^{-1} w$.

Proof. Since $A = A^*$, $(I + \mathrm{i}A)^{-1} = (I - \mathrm{i}A)(I + A^2)^{-1}$. Its Hermitian part is $(I + A^2)^{-1}$. Congruence by $G^{-1/2}$ proves (7), because $H = G^{1/2}(I + A^2) G^{1/2}$. The order inequalities follow from $0 \preceq A^2 \preceq \tau^2 I$. Completing the square at $v_* = H^{-1} w$ gives $Q - L(v) = (v - v_*)^* H (v - v_*) = r_H^* H^{-1} r_H$. Invert the order bounds in (7) to prove the residual interval. Strict positivity proves uniqueness of the maximizer. □

For the physical moments and the specific vector $w_j = \mathrm{i}^j$, the parity matrix $P = \mathrm{diag}((-1)^j)_{0 \leq j < N}$ satisfies $PGP = G$, $PJP = -J$, and $Pw = \overline{w}$. Thus $PMP = \overline{M}$, and conjugating $w^* M^{-1} w$ gives the same scalar after substitution of these relations. It is real. Therefore in the physical case $Q = w^* M^{-1} w > 0$ without merely taking its real part. This is not positivity of M as a Hermitian matrix. Nor do (7) imply cross-N monotonicity of H_N: the interior inverse G_N^{-1} changes with N.

Theorem 4.2 (Observable-adapted alternating hierarchy)**.** *Assume $0 \leq \tau < 1$ in Theorem 4.1, and $w \neq 0$. For $z = G^{-1/2} w$, define $\mu_j = z^* A^{2j} z \geq 0$ and $P_m = \sum_{j=0}^{m} (-1)^j \mu_j$. Then*

$$Q \geq \frac{\mu_0^2}{\mu_0 + \mu_1} \geq \frac{\mu_0}{1+\tau^2}, \tag{9}$$

$$Q - P_m = (-1)^{m+1} z^* A^{2m+2} (I + A^2)^{-1} z, \tag{10}$$

$$\frac{\mu_{m+1}}{1+\tau^2} \leq |Q - P_m| \leq \mu_{m+1}. \tag{11}$$

Consequently, for every integer $k \geq 0$,

$$P_{2k+1} \leq Q \leq P_{2k}, \quad P_{2k} - P_{2k+1} = \mu_{2k+1}, \quad \frac{P_{2k} - P_{2k+1}}{Q} \leq (1+\tau^2)\tau^{4k+2}. \tag{12}$$

All μ_j are computable without matrix square roots: starting with $c_0 = G^{-1} w$, recursively use

$$c_{j+1} = G^{-1} J c_j, \qquad \mu_j = c_j^* G c_j. \tag{13}$$

Proof. Cauchy–Schwarz for $(I+A^2)^{1/2}z$ and $(I+A^2)^{-1/2}z$ gives $\mu_0^2 \le (\mu_0+\mu_1)Q$. Since $\mu_1 \le \tau^2\mu_0$, (9) follows. The finite geometric-series identity for $B = A^2$ is

$$(I+B)^{-1} - \sum_{j=0}^{m}(-B)^j = (-1)^{m+1}B^{m+1}(I+B)^{-1}.$$

Apply it between z^* and z. As B commutes with its resolvent and $0 \preceq B \preceq \tau^2 I$, the magnitude is bounded by $\mu_{m+1}/(1+\tau^2)$ and μ_{m+1}, with the asserted sign. The two consecutive partial sums have difference μ_{2k+1}, which is at most $\tau^{4k+2}\mu_0$. Divide by the lower bound for Q in (9). Finally $G^{1/2}c_0 = z$, and the recursion gives $G^{1/2}c_j = A^j z$ inductively, proving (13). □

Here $\tau = \sqrt{2}-1$, uniformly in N. For $k = 3$ the analytic relative-width guarantee is

$$(1+\tau^2)\tau^{14} = 5.12754775030527608 \times 10^{-6} \quad \text{(decimal evaluation)}.$$

This is an arbitrarily refinable scalar certificate, not just a better fixed comparison factor. Each added step still requires a physical Gram solve and multiplication by J, and rounding errors must be enclosed. The uniform exact-arithmetic contraction is not a dimension-independent operation count or a proof of numerically stable monomial arithmetic.

Proposition 4.3 (Explicit reference-weight constants)**.** *Let $\lambda = \pi/8$, $c_\lambda = \cos\lambda$, and $S_N = \sum_{m<N}|p_m(3\mathrm{i}/8)|^2$ for polynomials orthonormal in $\nu(dv) = dv/\cosh(\pi v)$. Then*

$$\frac{3S_N}{8c_\lambda} \le Q_0 \le \frac{3S_N}{4c_\lambda}. \tag{14}$$

Conditionally on the source inverse identity, its deficit $D_N = 1 - c\sum_{m=1}^{N-1} h_N(m)$, $N \ge 3$, satisfies

$$\frac{3c_\lambda}{4}\frac{S_N}{N} \le D_N \le \frac{3}{2c_\lambda}\frac{S_N}{N}. \tag{15}$$

Proof. The ratio of $g(t)$ to $\operatorname{sech}(3\lambda t)$ is exactly

$$\frac{c_\lambda}{2}\left(1 + \frac{\cosh(2\lambda t)}{\cosh(4\lambda t)}\right),$$

between $c_\lambda/2$ and c_λ. Under $v = 3t/8$, the transformed physical measure is between $(4c_\lambda/3)\nu$ and $(8c_\lambda/3)\nu$. For any positive measure on this polynomial space the squared evaluation norm is $\sup_{f\neq 0}|f(3\mathrm{i}/8)|^2/\|f\|^2$. Comparing denominators gives (14). Now $\cos^2\lambda\, Q_0 \le Q \le Q_0$ by (7), and the source identity is $ND_N = 2Q$. Combining these facts gives (15). □

The explicit constants and alternating hierarchy do not change $D_N \asymp N^{-1/4}$. That asymptotic still needs the classical polynomial evaluation estimate and the source identification of D_N.

5 Certified finite solves in the physical metric

The purpose here is an error certificate for inexact realizations of the preceding exact constructions in the source's accretive polynomial problem. Write $M = G + \mathrm{i}J$ with the actual G_N, J_N above, and $\|v\|_{G^{-1}} = (v^*G^{-1}v)^{1/2}$. In this section $Q = w^*M^{-1}w$ denotes the full, possibly complex scalar for arbitrary w; Section 4 used Q for its real part.

Theorem 5.1 (Residual and perturbation certificates)**.** *For any $N \geq 1$, $w \in \mathbb{C}^N$, and approximate coefficient vector $\widehat{u}$, set $r = w - M\widehat{u}$, $u = M^{-1}w$, $Q = w^*u$, and $\widehat{Q} = w^*\widehat{u}$. Then*

$$\|u - \widehat{u}\|_G \leq \|r\|_{G^{-1}}, \tag{16}$$

$$|Q - \widehat{Q}| \leq \|w\|_{G^{-1}} \|r\|_{G^{-1}}. \tag{17}$$

If $\widehat{M} = M + E$ and $\|G^{-1/2}EG^{-1/2}\| \leq \delta < 1$, then $\widehat{M}$ is invertible and

$$|w^*\widehat{M}^{-1}w - w^*M^{-1}w| \leq \frac{\delta}{1-\delta}\, w^*G^{-1}w. \tag{18}$$

For $M(t)$ differentiable at a real parameter value, with $M'(t) = E$ and that value's positive metric G, one also has

$$|\partial_t(w^*M(t)^{-1}w)| \leq \|G^{-1/2}EG^{-1/2}\|\, w^*G^{-1}w \tag{19}$$

when w is fixed; variable covectors require their two derivative terms.

Proof. Set $A = G^{-1/2}JG^{-1/2} = A^*$ and $C = I + \mathrm{i}A$. For every v, $\|Cv\|^2 = \|v\|^2 + \|Av\|^2$, so $\|C^{-1}\| \leq 1$ and $M^{-1} = G^{-1/2}C^{-1}G^{-1/2}$. Apply this to $u - \widehat{u} = M^{-1}r$ and then use Cauchy–Schwarz between $G^{-1/2}w$ and $G^{1/2}(u - \widehat{u})$. For $D = G^{-1/2}EG^{-1/2}$, a Neumann series for $I + C^{-1}D$ gives $\|(C+D)^{-1}\| \leq (1-\delta)^{-1}$. The resolvent identity $(C+D)^{-1} - C^{-1} = -(C+D)^{-1}DC^{-1}$ proves (18). Finally differentiate the inverse and bound both copies of C^{-1} by one to prove (19). □

For a computed Gram approximation satisfying $aG \preceq \widehat{G} \preceq bG$, $0 < a \leq b$, one can replace the uncomputed dual norms in (17) by

$$|Q - \widehat{Q}| \leq b\|w\|_{\widehat{G}^{-1}} \|r\|_{\widehat{G}^{-1}}.$$

Indeed $G^{-1} \preceq b\widehat{G}^{-1}$. A computed residual must also include data error: if $\widehat{r} = w - \widehat{M}\widehat{u}$, then $r = \widehat{r} + E\widehat{u}$ and $\|r\|_{G^{-1}} \leq \|\widehat{r}\|_{G^{-1}} + \delta\|\widehat{u}\|_G$. Entrywise tolerances without this relative metric control are not certificates. Positive quadrature on fewer than N nodes has rank less than N and cannot certify $G > 0$ on X_N.

The source bound $\|A\| \leq \tan\lambda$ also gives $\Re Q \geq \cos^2\lambda\, w^*G^{-1}w$. Thus (18) supplies relative scalar error at most $\delta/((1-\delta)\cos^2\lambda)$ relative to $\Re Q$. This explicitly states what a dimension-uniform finite computation would have to establish before it could preserve a power exponent. It does not itself supply a bound on δ for growing N. An additional enclosure avoids cancellation in the scalar pairing. For $u = M^{-1}w$, taking real parts of $w^*u = (Mu)^*u$ gives $\Re Q = u^*Gu$. With an approximate $\widehat{u}$ and the true residual r, the triangle inequality and (16) imply

$$\max(0, \|\widehat{u}\|_G - \|r\|_{G^{-1}})^2 \leq \Re Q \leq (\|\widehat{u}\|_G + \|r\|_{G^{-1}})^2.$$

For the specific physical parity vector, Q is real and these also enclose the entire pairing; that stronger conclusion is not asserted for arbitrary complex vectors.

6 A normalization-sensitive sewing improvement

This section isolates a local gain in the spatial argument. It does not assume that algebraic exactness supplies disjoint paths. All geometric recovery and chord estimates are stated as hypotheses.

6.1 The weighted representation inequality

For a countable set of paths with positive weights w_ξ, put $E_1 = \sum w_\xi m(\xi)$ and $E_2 = \sum w_\xi m(\xi)^2$. If $0 < E_2 < \infty$, then

$$\sum_{m(\xi)>0} w_\xi \geq E_1^2/E_2. \tag{20}$$

For finite subsets this is Cauchy–Schwarz applied to $\sqrt{w_\xi}$ and $\sqrt{w_\xi} m(\xi)$; increasing finite subsets prove the countable case. It applies to noninteger thinning-weight multiplicities. This is classical, not a new Ram inequality. Its source-specific use requires recoverability of cuts, not merely disjointness at construction.

For an interval of L integer levels define

$$S_q(L) = \sum_{d,d'=1}^{L} (1 + |d - d'|)^{-q} = L + 2\sum_{h=1}^{L-1} (L-h)(1+h)^{-q}.$$

Integral comparison, and the restriction $h \leq L/2$ for lower bounds, give

$$S_q(L) \asymp \begin{cases} L^{2-q}, & 0 < q < 1, \\ L\log(L+1), & q = 1, \\ L, & q > 1. \end{cases} \tag{21}$$

Fixed grid spacings and windows of comparable length change constants only.

Theorem 6.1 (Two-arm sewing with the actual thinned mass)**.** *Fix $0 < p < 1$, $z \in \mathbb{R}$, and three independent admissible offset windows of length comparable to R. Suppose for a bulk vertex v that:*

1. *retained ordered central chords, with a fixed ordered side pair, have total mass $\mathcal{S}_R \geq R^{3+z-\eta_R}$, $\eta_R \to 0$;*
2. *each role's arm family depends only on its triangle and own initial port, lies in one fixed corridor, and has been thinned to exactly $a_R > 0$ mass, uniformly over those data;*
3. *for any two triangles representing one resulting bridge through v, the recovered common marked chord has mass at most $U_R \leq R^{z+\eta_R}$; each intervening gap bridge has total mass at most $C(1+\Delta)^{-p}$; the exterior complete arms are from the recovered roles and own initial ports; all remaining thinning factors are at most one;*
4. *the valid central interval is uniquely recoverable at each fixed triangle and the concatenated paths are simple bridges.*

Then the mass $W_R(v)$ of distinct resulting bridges through v obeys

$$W_R(v) \gtrsim a_R^2 R^{z+2p-3\eta_R} \quad (0 < p < 1/2). \tag{22}$$

At $p = 1/2$ the same bound holds with an additional divisor $\log(R+1)$; for $1/2 < p < 1$ a bound valid for both side cases is

$$W_R(v) \gtrsim a_R^2 R^{z+1-3\eta_R}. \tag{23}$$

If only different-side chords are used, (22) holds for every $0 < p < 1$ without the logarithmic divisor.

Proof. Let $m_D(\xi)$ be the product of the two thinning factors at triangle D and $m = \sum_D m_D$. Independence of each arm normalization from the opposite endpoint and disjoint concatenation give $E_1 = \mathcal{S}_R a_R^2$. For fixed D, D', retain the factors for the arm selected at exterior entry and the arm selected at exterior exit. If their representation labels differ, the own-port condition still identifies both normalized families. At ties make a fixed choice; even if $D = D'$, the unused copy of each thinning factor is at most one. Recovery and removal of other avoidance restrictions therefore give

$$\sum_{\xi} w_\xi m_D(\xi) m_{D'}(\xi) \le C U_R a_R^2 (1+\Delta_A)^{-p} (1+\Delta_B)^{-p}.$$

For different sides, the two used offset pairs contribute $S_p(O(R))^2 = O(R^{4-2p})$ and the unused pair contributes $O(R^2)$. Thus $E_2 \lesssim U_R a_R^2 R^{6-2p}$. For the same side the two gaps are equal; the bound is instead $E_2 \lesssim U_R a_R^2 R^4 S_{2p}(O(R))$. Apply (21) and (20). Since $\mathcal{S}_R^2/U_R \ge R^{6+z-3\eta_R}$, these are precisely the asserted bounds. No path count was replaced by a representation count. □

Corollary 6.2 (Retaining the source's arm-loss factor)**.** *Under the source's stated recovery and separated-supply estimates, with $p = 1/4$, $z = 1/12$, and $a_R = R^{-p-K_{\mathrm{loss}}\epsilon}$, the local distinct-bridge bound improves from the source's displayed $R^{z-4K_{\mathrm{loss}}\epsilon-o(1)}$ to*

$$W_R(v) \ge R^{z-2K_{\mathrm{loss}}\epsilon-o(1)}.$$

Proof. Use (22). The improvement is obtained by retaining a_R^2, rather than the larger bound R^{-2p}, for the two recovered thinned arms in the second moment. All source geometric hypotheses are unchanged; the first moment already used the actual normalization. □

This gain makes exponent-loss accounting sharper at each fixed ϵ. It changes neither the limiting $4/3$ exponent nor the corridor width, and no numerical speedup is claimed. The threshold statement is a conditional extension of the counting mechanism, not a claim that a honeycomb model with arbitrary p exists or that its other source identities hold unchanged.

6.2 A conditional mean theorem and fixed-corridor quantifiers

Suppose, independently, that the source geometric hypotheses above hold in one corridor $C_0 R$ for every fixed $\epsilon > 0$, with K_{loss} independent of ϵ, that the translation counting has $\asymp R^2$ bulk vertices and $O(R)$ starting ports over finitely many lattice-congruent strip choices, and that

$$M(R) \le R^{1+z+o(1)}, \qquad B_C(R) \asymp R^{-p} \quad (C \ge C_0 \text{ sufficiently large and fixed}).$$

Then for $0 < p < 1/2$ Theorem 6.1 gives

$$M_C(R) \ge R^{1+z-2K_{\mathrm{loss}}\epsilon-o(1)}.$$

For any target slack $\delta > 0$, choose one fixed $\epsilon < \delta/(4K_{\mathrm{loss}})$ when $K_{\mathrm{loss}} > 0$, and then take R sufficiently large for that choice; the remaining $o(1)$ is below $\delta/2$. If $K_{\mathrm{loss}} = 0$, take any allowed fixed ϵ. Hence $\liminf \log M_C / \log R \ge 1 + z$ with the *same* corridor. The assumed upper bound gives equality, and division by B_C gives exponent $1 + z + p$. There is no simultaneous limit $\epsilon = \epsilon(R)$ justified by this argument. The source's constant corridor is stronger than a corridor whose width grows like $R \log^2 R$.

7 Comparison with primary public literature

Public searches on October 7, 2026 used only public mathematical titles and topics. No private passages, file contents, or confidential report text were placed in queries. The following are selected primary comparisons, not a claim of an exhaustive literature search.

Duminil-Copin and Smirnov prove the exact honeycomb connective constant using the parafermionic observable [1]. This supports the critical weight and classical exploration background, not the claimed bridge mean or sharp arch power. Dyhr, Gilbert, Kennedy, Lawler and Passon study the critical strip law with one endpoint summed and its relation to conditioning a half-plane law [2]. The ensemble comparison is pertinent, but it does not identify this annealed length exponent with a fixed-length displacement theorem.

Krachun and Panagiotis' article is published in *Annals of Probability* 54(3), 1109–1125 (2026), rather than only a future or unpublished item [3]. Its primary result gives quantitative sub-ballisticity, with high-probability radius $O(n/\log n)$, and a polynomial decay bound for critical bridge mass. Those claims are weaker and differently conditioned than the sharp powers of M460; we do not credit them with establishing M460's assertions.

Glazman and Manolescu give Yang–Baxter two-point invariance for their specified rhombic model and a proof of bridge-mass decay [4]. That integrable framework is an antecedent of the source's local exchange calculation, not permission to change the physical measure arbitrarily. Lawler, Schramm and Werner's scaling-limit discussion identifies the conjectural $\mathrm{SLE}_{8/3}$ description under conformal-invariance assumptions [5]; it is not a completed lattice proof of the present theorem.

The Meixner–Pollaczek generating function and orthogonality belong to classical special-function theory, explicitly recorded in DLMF §§18.19 and 18.23 [6]. The source-specific use here is the evaluation point 3i/8 and the physical inverse pairing, not the discovery of the polynomial family. Likewise Gram whitening, accretive resolvent estimates, exact moment recurrences, and weighted Cauchy–Schwarz are classical techniques. Private Ram representations organize the coordinates, metric, and error account. The realized gains in this report are the explicit source-specific certificates and the normalization-sensitive conditional sewing result, without a claim of historical priority for their constituent methods.

8 Verification, remaining dependencies, and assessment

8.1 Verification contract

The companion `_audit/M460-author-input-ledger.json` records source, private-input, policy, helper, and parent-evidence SHA-256 hashes, read ranges, and public URLs. The owned check implementation `_audit/M460-author-checks.py` produced `_audit/M460-author-checks.json` with status PASS. Its exact checks at $N = 1, 2, 3, 4$ cover the complex congruence with the transported covector, the Ram moment-coordinate bank, effective-Gram inverse, parity reality, and $Q = u^*Gu$. The moment-polynomial parity recurrence was checked through degree ten. Its seeded inexact-solve diagnostics at $N = 3, 4, 5$ satisfy both physical residual bounds, and 16 finite offset-sum cases agree with the exact one-dimensional summation formula. Symbolic algebra checks the old and improved sewing exponents and $13/12 + 1/4 = 4/3$.

The first independent matrix implementation and output, `_audit/M460-independent-matrix.py` and `_audit/M460-independent-matrix.json`, were read without alteration. They verify effective-Gram inverse, parity reality, and variational residual identities exactly for $N = 3, 4, 5, 6$,

and check the alternating and observable-adapted inequalities numerically. They give

$$Q_3 = \frac{6}{7} + \frac{3\sqrt{2}}{14}, \qquad Q_4 = \frac{19}{23} + \frac{10\sqrt{2}}{23}.$$

That independent calculation's decimal values of Q_N are

N	Q_N (rounded)
3	1.16018862050852037
4	1.44096241842308480
5	1.70422794408529428
6	1.95441958825947652

The author independently reproduces the exact $N = 3$ value. The all-N inequalities are proved in this report; the decimal checks are diagnostics, not interval-certified proofs or evidence for an asymptotic exponent. Neither implementation enumerates walks or replays the full braid, interpolation, contour, or geometric argument.

A separate audit, `_audit/M460-secondary-audit.txt`, reconstructs the moments by reciprocal series for $\cos(\lambda + u)/\cos\lambda$ instead of the derivative-polynomial recurrence. It records exact checks through $N = 6$ of brackets $k = 0, 1, 2, 3$, remainders $m = 0, \ldots, 7$, width bounds, Rayleigh inequalities, and real-trial residual identities and inequalities. Signs of elements $a + b\sqrt{2}$ are decided by rational sign and square comparisons. These are exact-sign checks, unlike the first independent script's floating-point sign tests of $P_3 \le Q \le P_2$. The separate audit also records a 70-digit direct-density quadrature cross-check for moment indices 0 through 18, with maximum scaled discrepancy $2.3282569 \times 10^{-70}$. This quadrature was not interval-certified. Its ephemeral checking program was not saved; the evidence artifact is the signed-scope audit text, not a claim that the author reran an unavailable implementation. The all-N analytic proofs, constants, parity sign, and normalization were independently reviewed with no substantive error found. Review clarified $w \neq 0$ for relative bounds and corrected the general-vector energy enclosure to use $\Re Q$; both qualifications are incorporated above.

Two failed author-check setup attempts are recorded: the bundled Python lacked SymPy, and the pre-existing Anaconda SymPy required an explicit field conversion for the $N = 1$ rational matrix. After that checker-only correction, all listed tests passed using the existing Anaconda Python with SymPy 1.11.1 and NumPy 1.23.5. No package was installed. Disk PDF export uses the mandated bundled Python and the existing MiKTeX through `_tools/compile_report.py`, not that test runtime.

The coordinated native-editor attempt reported the source open as queued and the actual built-in compiler status as `compile-failed`, with message "Unable to find standard directories for platform" followed by a newline. Native compilation is therefore not claimed successful. The first disk export succeeded with 17 pages and no undefined references; all-page contact inspection identified three minor source-label overflows, corrected in the final source. Mathematical review was incorporated before the second and last format QA round. The final freeze manifest is the authoritative record of final source, PDF, check, review, and input hashes.

8.2 Unmet dependencies and priorities

1. The rapidity-orbit construction, scalar commutant, and complete diagram identities are read source arguments, not independently replayed here. Numerical inverse tests cannot substitute for them.

2. The infinite oscillator/current and residue analysis requires a specialist review of convergence, zero modes, continuation, and the uniform complex-angle bound. It is the largest unreplayed analytic dependency in the cylinder exponent.

3. The determinant normalization needs the projection kernel's uniform bounds and off-subspace Hilbert–Schmidt estimate. The private physical-metric correspondence diagnoses this need but does not fill it.

4. The marked three-leg identity, filled-domain deletion, fixed-end plateau, and path-order recovery must be checked geometrically. The conditional sewing theorem assumes these steps explicitly.

5. No effective constants for all source asymptotics, no amplitude limit, no scaling-limit theorem, no fixed-length exponent, and no distributional concentration of bridge length are established here.

8.3 Advantages, costs, and future research

Exact derivative moments avoid an arbitrary truncation of the real line. Metric-relative certificates expose whether a finite solve concerns the same observable after a basis change. Own-port normalization keeps the two-representation argument valid when exterior arms come from different triangles. These are useful, auditable improvements in representation and inference.

Monomial matrices can be badly conditioned even when the physical operator is uniformly accretive. Exact symbolic derivatives can grow rapidly in size; no favorable complexity bound is asserted. Whitening does not reduce the dimension of the polynomial space or replace the asymptotic evaluation estimate. The sharpened arm-loss coefficient affects finite-slack bounds, not the headline exponent.

Promising next steps are certified orthogonal-coordinate moment arithmetic with relative Gram errors; targeted formal checks of the small fusion compatibilities; an independent proof of the reflected contour stability estimates; and a geometric verification of mixed entry/exit recovery. If better same-side endpoint selection were proved, the $p = 1/2$ threshold in the abstract conversion could potentially be avoided, but that is not obtained by the current proof.

8.4 Realized contribution

The evidence supports C3, not an automatic consequence of L5 relevance. Proposition 3.1, Theorems 4.1, 4.2, 5.1, and 6.1 give exact local constructions and proved conditional conversion gains; the full-source audit makes the missing analytic and spatial inputs explicit. This is a notable verification and representation contribution relative to the pinned manuscript. It is not C4 solely because an exponent-loss coefficient is halved, and it is not an independent proof of the sharp critical exponents.

References

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